



**QI26 - 20: Quantum Computing for Combinatorial Assignment Optimization**

# A Reproducible QUBO–QAOA Benchmark for Generalized Bipartite Assignment on NISQ Hardware

---

**Verified progress · Real-hardware evidence · Preliminary novelty assessment**

**Research team:** Iván Barrientos Salas · Ilia Fazeli · Jiya Maurya · Marcos Jiménez · Abhishek Raj · Satkar Juneja

**PHASE 1 · EXECUTABLE ARTIFACT**

# What the Project now demonstrates

Five claims supported by the executed artifact and the archived IBM Quantum job.

## 1 GM-QAOA works – ideally

Feasible probability = 1.0 and state-preparation / full-circuit fidelity = 1.0 on all three fixed instances.

## 2 Standard QAOA is functional

It recovers optimal scores in multiple settings; on the 4×2 capacity instance, feasible mass rises to 43.99% at p=3.

## 3 The pipeline reaches a QPU

Both p=1 circuits ran in one ibm\_marrakesh Runtime job and sampled the exact optimum (score 3.25).

## 4 Rigor and traceability are built in

Same instances, score, and decoder; exact and Hungarian ground truth; exported CSV/JSON; fixed seeds; archived job ID; claims checklist and compliance matrix.

## 5 A preliminary originality result

Within the searched corpus, no prior publication was found that combines exact-capacity assignment, standard vs. GM-QAOA, controlled same-job hardware execution, full-distribution errors, and transpilation-resource analysis.

### THE DEFENSIBLE MESSAGE

**There is real, measurable progress: the algorithms function, the workflow is reproducible, and the experiment reaches present-day quantum hardware.**

# Generalized bipartite assignment as QUBO

One mathematical model, three controlled instances, one common scoring and validation layer.

## DECISION VARIABLES

$x_{ij} \in \{0,1\}$  · agent  $i \rightarrow$  resource  $j$

## CONSTRAINTS

- Each agent is assigned exactly once.
- Resource use is either one-to-one or fixed by capacity vector  $C$ .
- All classical and quantum workflows use the same decoder.

## IMPLEMENTED QUBO

$$Q(x) = - \sum_{ij} S_i x_{ij} + \lambda_A \sum_i (\sum_j x_{ij} - 1)^2 + \lambda_R \sum_j (\sum_i x_{ij} - C_j)^2$$

Penalty coefficients are generated programmatically; the QUBO-to-Ising mapping is numerically validated.

reference 3×3

**9 qubits · 6 feasible**

One-to-one assignment · optimum score 2.70

one to one 4×4

**16 qubits · 24 feasible**

Reproducible seed-2026 instance · optimum score 3.08

exact capacity 4×2

**8 qubits · 6 feasible**

Capacity vector  $C=(2,2)$  · optimum score 3.25

# Seven-stage reproducible research Pipeline

The notebook is an executable artifact, not a collection of disconnected code cells.



## CONTROLLED COMPARISON

**same instances + same score + same feasibility decoder + same backend protocol**

### Ground truth

Exact optimum and classical gaps

### Algorithm quality

Feasible / optimal probabilities

### Hardware quality

Counts, TVD, Hellinger, fidelity

### Physical cost

Depth, instructions, native CZ

# Strong classical reference points were established

The quantum results are interpreted against exact optima—not weak or ambiguous baselines.

**3×3 one-to-one**

**2.70**

GLOBAL OPTIMUM

Exact · Hungarian · Greedy · SA  
all recover the optimum.

**4×4 one-to-one**

**3.08**

GLOBAL OPTIMUM

Exact and Hungarian are optimal.  
Greedy = 2.80 · SA = 3.01.

**4×2 exact capacity**

**3.25**

GLOBAL OPTIMUM

Exact · Greedy · SA  
all recover the optimum.

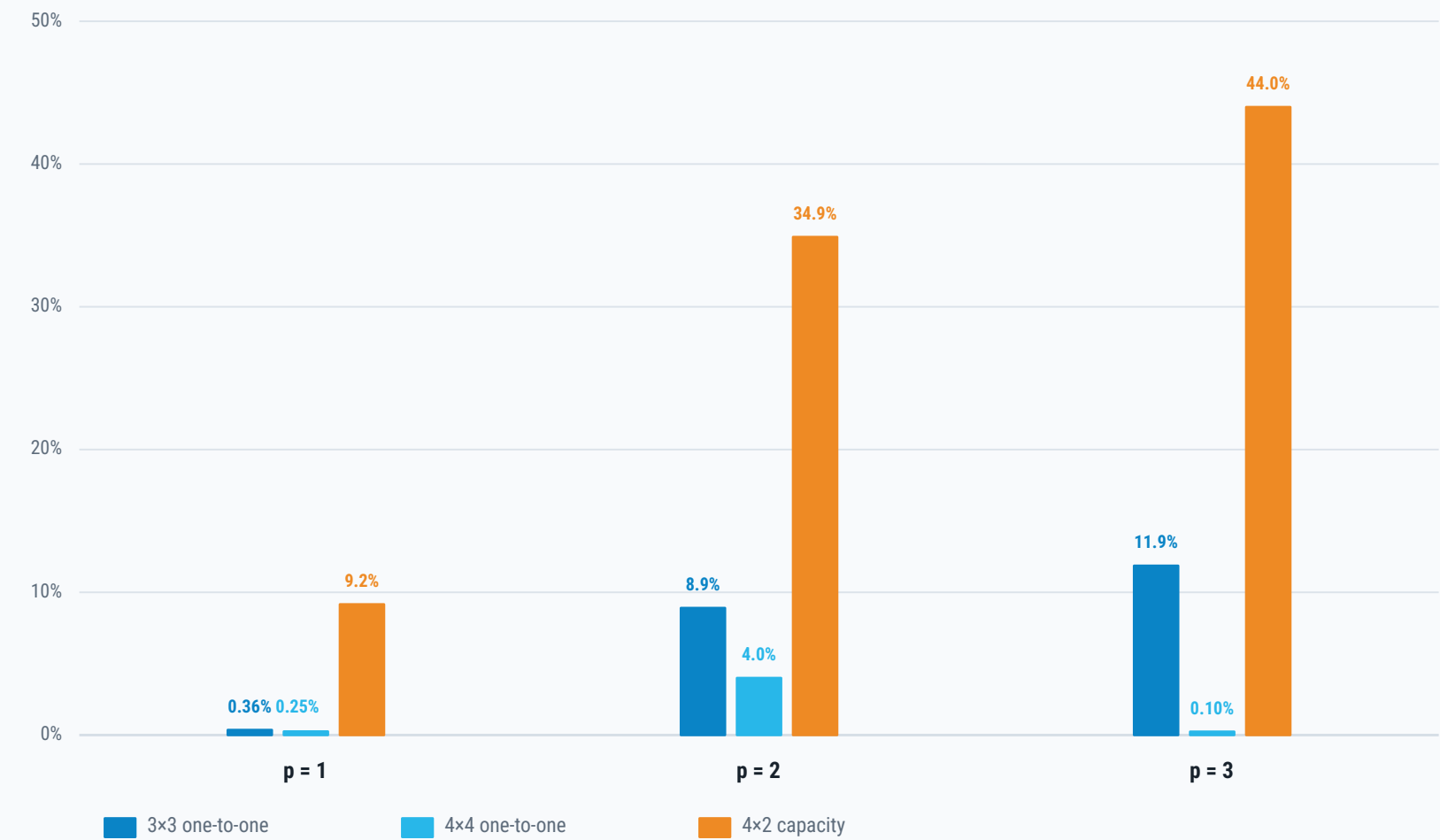
## SCIENTIFIC ROLE OF THE BASELINES

**They establish the correct optimum, expose non trivial heuristic failure on 4×4, and make zero-gap quantum samples interpretable.**

**Scope boundary: at these sizes, exact and Hungarian solvers are much faster.  
This is a correctness and NISQ implementation benchmark—not a runtime competition.**

# Standard QAOA is functional in ideal simulation

Feasible probability by depth  $p$ ; labels show measured feasibility, while the right panel highlights the strongest tested settings.



3x3 ·  $p=3$

**PF 11.86%**

Popt 4.15% · best score 2.70

4x4 ·  $p=2$

**Optimum 3.08**

Only tested depth with zero gap

4x2 ·  $p=3$

**PF 43.99%**

Popt 6.11% · best score 3.25

**Functional  $\neq$  monotonic with depth  $p$**



# GM-QAOA works as an ideal feasible-subspace Ansatz

The strongest verified result of Phase 1: feasibility and state-preparation fidelity are exactly preserved in ideal simulation.

## reference 3×3

**Popt 69.2%**

PF = 100%  
F(US) = 1.0  
Circuit fidelity = 1.0  
Best score = 2.70  
Ideal depth = 1,062

## one to one 4×4

**Popt 10.7%**

PF = 100%  
F(US) = 1.0  
Circuit fidelity = 1.0  
Best score = 3.08  
Ideal depth = 20,252

## exact capacity 4×2

**Popt 50.4%**

PF = 100%  
F(US) = 1.0  
Circuit fidelity = 1.0  
Best score = 3.25  
Ideal depth = 395

### WHAT IS PROVED

**The implemented preparation + Grover mixer preserve the fixed feasible subspaces exactly.**

### SCOPE

**Exact fixed-size constructions; not a general scalable library.**

# The end-to-end pipeline reached real hardware

A controlled comparison: same backend, same job, same shots, same pass manager, same hardware options.

ibm\_marrakesh · p=1 · 1,024 shots per circuit · optimization level 3 · dynamical decoupling · gate twirling

## STANDARD QAOA

**3.25**

best feasible score  
exact optimum · gap 0.00

Observed probability

**PF 7.71%**

**Popt 1.56%**

Physical resources

**Depth 143**

**CZ 63**

## GM-QAOA

**3.25**

best feasible score  
exact optimum · gap 0.00

Observed probability

**PF 11.91%**

**Popt 2.83%**

Physical resources

**Depth 1,357**

**CZ 496**

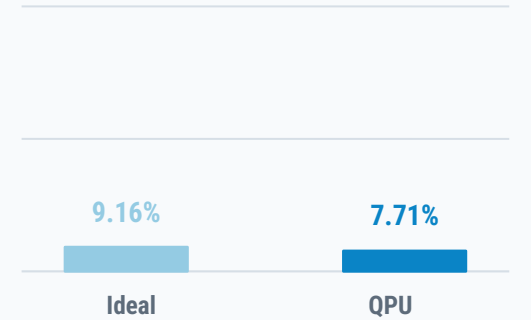
Both hardware distributions contained the exact optimum bitstring **10011001** · Runtime job d9vl8qfo3ppc73akfak0



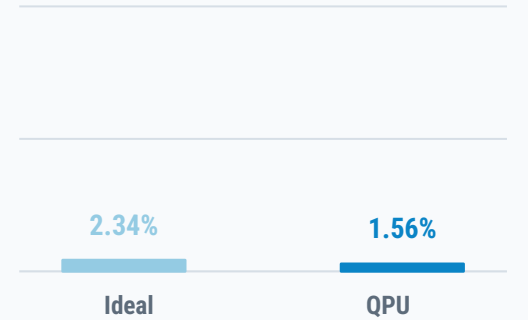
# Best-sample success is real, but it is not statistical reliability

The optimum appeared in both distributions, while ideal-to-hardware concentration degraded—especially for GM-QAOA.

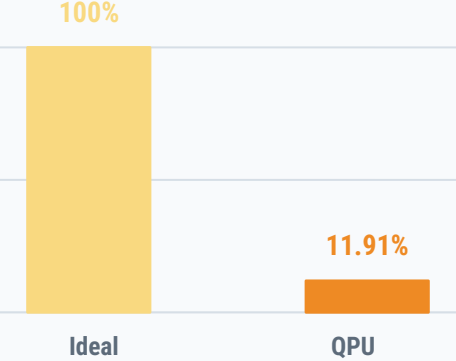
Standard QAOA · feasible probability



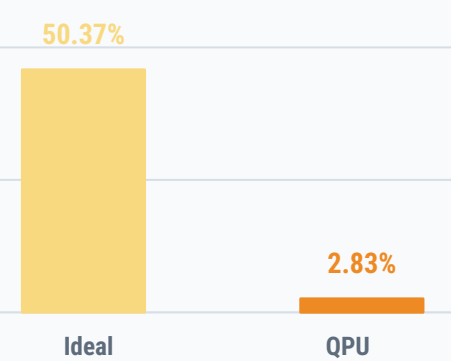
Standard QAOA · optimal probability



GM-QAOA · feasible probability



GM-QAOA · optimal probability



STANDARD QAOA

**16 / 1,024** optimal-state counts

Its ideal optimum probability lies inside the physical Wilson interval.

GM-QAOA

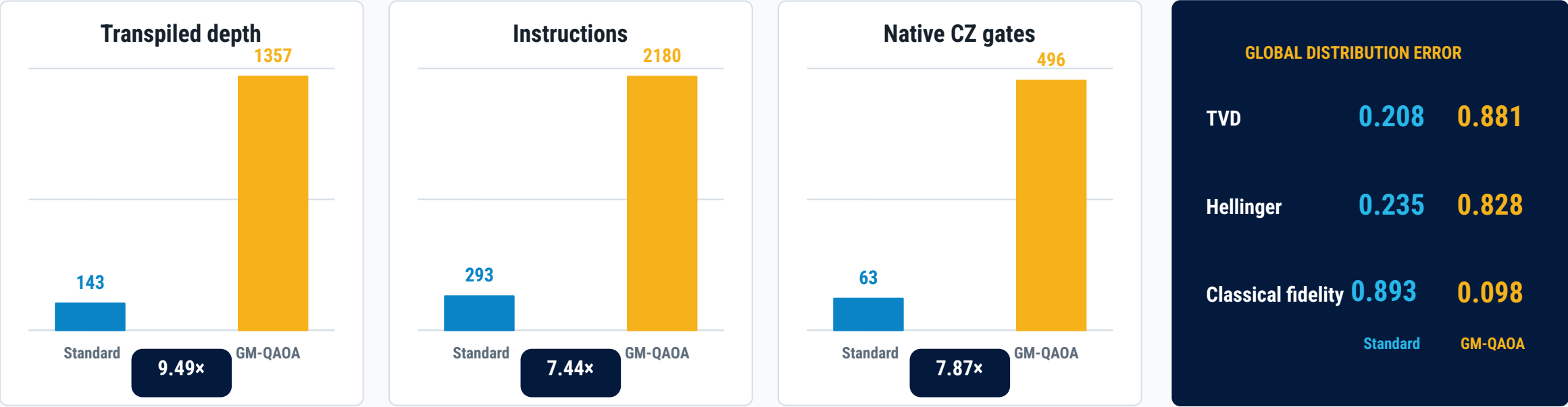
**29 / 1,024** optimal-state counts

Its ideal 50.37% optimum probability is far outside the physical interval.

Correct interpretation: “the optimum was sampled” — not “the optimum was found with high probability.”

# The ideal GM-QAOA advantage collapses under physical overhead

The data strongly support circuit length and synthesis/routing cost as major explanatory factors—without proving depth alone is causal.



PRIMARY SCIENTIFIC INTERPRETATION

GM-QAOA does not fail as an ideal ansatz. The specific fixed state preparation and selective-phase implementation are too costly to preserve the ideal advantage on this hardware execution.

# A combination-level novelty result—stated cautiously

The individual ingredients have precedents; the searched corpus did not reveal the same controlled experimental combination.

## THE EXPERIMENTAL INTERSECTION



***“To the extent of the literature search conducted, this work presents an experimental combination not previously reported.”***

Narrow claim: novelty lies in the controlled combination—not in inventing QAOA, GM-QAOA, QUBO assignment, IBM hardware execution, or the individual metrics.

## CLOSEST PRECEDENTS

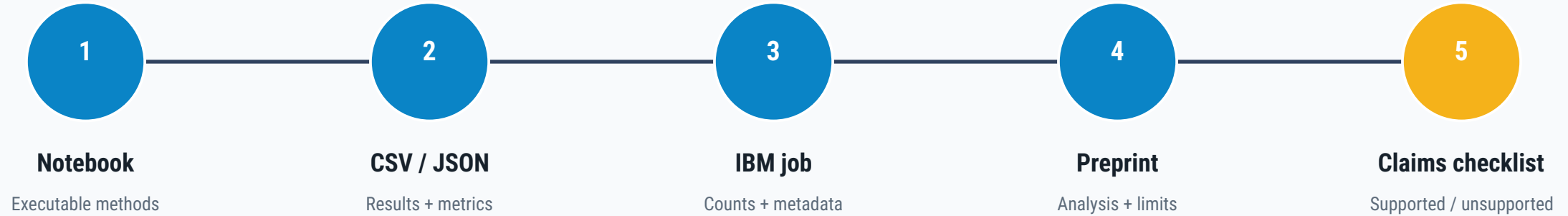
- Rhodes et al. (2025) – constrained vs. penalty mixer on IBM hardware
- Bärtschi & Eidenbenz (2020) – GM-QAOA method
- Stollenwerk et al. (2020) – hard-constrained assignment / resource estimates
- Golden et al. (2023) – Grover vs. transverse mixer scaling
- Hadfield et al. (2019) – alternating-operator framework

## SEARCH CAVEAT

36 full-text articles were reviewed, but coverage was heterogeneous and not exhaustive of arXiv, IEEE Xplore, or Scopus. The conclusion remains provisional.

# Evidence over hype: Every claim has an artifact boundary

This evidence-first framing aligns the presentation with QWorld's scientific-integrity principles.



## SUPPORTED BY THE ARTIFACT

- One-to-one and fixed exact-capacity QUBOs are implemented and validated.
- Standard QAOA is compared at  $p=1,2,3$  in ideal simulation.
- Fixed GM-QAOA circuits preserve the feasible subspace ideally.
- Both physical circuits sampled the exact optimum.
- The recorded GM-QAOA circuit has much higher depth and CZ count.

## NOT SUPPORTED BY THE ARTIFACT

- Quantum advantage or practical runtime superiority.
- High-probability optimum recovery on hardware.
- QPU preservation of the GM-QAOA feasible subspace.
- A general scalable state-preparation construction.
- Universal overhead ratios across devices or instances.

# Phase-1 Conclusions and the next research decisions

A balanced summary: verified achievements first, then the experiments required to generalize them.

## 1 GM-QAOA works ideally

Exact fixed-instance preparations preserve feasibility and produce much higher  $p=1$  optimum probabilities.

## 2 Standard QAOA works

It recovers optimal scores in multiple settings; depth effects are instance- and metric-dependent.

## 3 The hardware pipeline works

Both matched  $p=1$  circuits executed on `ibm_marrakesh` and sampled the optimum.

## 4 The physical trade-off is measurable

GM-QAOA's state-preparation/mixer cost destroys most of its ideal distribution advantage.

## 5 The combination appears underreported

No exact methodological match was found in the searched corpus; this remains a cautious, provisional claim.

**NO QUANTUM-ADVANTAGE CLAIM**

**The contribution is a reproducible formulation, a controlled hardware experiment, and evidence-based error/resource analysis.**

### NEXT RESEARCH DECISIONS

**01 Repeat matched jobs across time and multiple backends.**

**02 Replace generic fixed preparation with structured scalable circuits.**

**03 Run mitigation and ablation controls for DD, twirling, readout, and postselection.**

**04 Construct controlled size/capacity families and report total resource cost.**



## THE TAKEAWAY

**The feasible-subspace construction works.  
Standard QAOA works.  
The complete pipeline reaches real hardware.  
The evidence is reproducible and bounded.**

*Within the searched corpus, the controlled experimental combination was not previously reported.*

No quantum advantage is claimed.  
Phase 1 establishes a transparent foundation for the next experimental campaign.

<https://github.com/qubitmx/qi26-20-quantum-assignment-optimization>

PHASE 1

**COMPLETE**

QI26-20  
QUBO · QAOA · GM-QAOA  
IBM QUANTUM

THANK YOU!